

Faculty of Science, Engineering and Built Environment

SEE312 Data Communication

Deakin University Unit Guide

Trimester 1, 2025

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Welcome

The unit provides an understanding in the basic principles of data communications and computer networking. The fundamental aspects underpinning the core developments in state of the art technologies, operational philosophies and hierarchical architectures in this field of modern computer networks and internetworking, will be examined in detail. In particular, the unit will cover the TCP/IP protocol suite with an emphasis on layers architecture, transmission mediums, data encoding, error detection and correction, medium access control, local and wide area networks as well as ATMS, Cellular Wireless Networks, bridges and routing technologies, node-to-node communications and end-to-end protocols.
This Unit Guide provides you with the key information about this unit. Please read it carefully and refer to it frequently throughout the study period. Your unit site also provides information about your rights and responsibilities. We will assume you have read this before the unit commences, and we expect you to refer to it throughout the study period.

To be successful in this unit, you must:

- read all materials in preparation for your learning activities and follow up each with further study and research on the topic
- start your assessment tasks well ahead of the due date
- read or listen to all feedback carefully and use it in your future work
- attend and engage in all educator facilitated (scheduled) learning activities and other learning experiences as part of the unit design

Who is the unit team?

Unit chair: leads the teaching team and is responsible for overall delivery of this unit

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Other members of the team and how to contact them

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Administrative queries

- check-out the 'SEBE Student Hub' section on your unit site
- contact your Unit Chair or Campus Leader
- drop in or contact Student Central to speak with a Student Adviser

For additional support information, please see the Rights and Responsibilities section under 'Content' in your unit site.
About this unit

Unit development in response to student feedback

Every trimester, we ask students to tell us, through eVALUate, what helped and hindered their learning in each Unit. You are strongly encouraged to provide constructive feedback for this Unit when eVALUate opens (you will be emailed a link).
In previous versions of this unit, students have told us that these aspects of the Unit have helped them to achieve the learning outcomes:

- organisation of the unit
- high quality of the unit material

The following aspects of the unit have been introduced, enhanced or retained in response to feedback from students who have undertaken this unit in previous trimesters:

- More detailed insight will be provided into how the End-of-Unit Assessment is expected to capture the student knowledge and understanding pertaining to the unit material.

If you have any concerns about the Unit during the trimester, please contact the unit teaching team - preferably early in the trimester - so we can discuss your concerns, and make adjustments, if appropriate.

Learning Outcomes

Each Unit in your course is a building block towards Deakin's Graduate Learning Outcomes - not all Units develop and assess every Graduate Learning Outcome (GLO).

ULO	These are the Unit Learning Outcomes (ULOs) for this unit. At the completion of this unit, successful students can:	Alignment to Deakin Graduate Learning Outcomes (GLOs)
ULO1	Understand the basic hardware and software environments for data communications.	GLO1: Discipline-specific knowledge and capabilities GLO2: Communication GLO3: Digital literacy GLO5: Problem solving GLO6: Self-management
ULO2	Explain the need for protocols and describe the current networking paradigms at a theoretical level.	GLO1: Discipline-specific knowledge and capabilities GLO2: Communication GLO3: Digital literacy GLO6: Self-management
ULO3	Describe various types of computer networks.	GLO1: Discipline-specific knowledge and capabilities GLO2: Communication GLO3: Digital literacy GLO4: Critical thinking
ULO4	Understand the advanced high-speed multimedia network technologies.	GLO1: Discipline-specific knowledge and capabilities GLO2: Communication GLO3: Digital literacy GLO6: Self-management
ULO5	Describe the TCP/IP protocol architecture.	GLO1: Discipline-specific knowledge and capabilities GLO2: Communication GLO3: Digital literacy GLO5: Problem solving GLO6: Self-management

Assessing your achievement of the unit learning outcomes

Hurdle requirements

To be eligible to obtain a pass in this unit, students must achieve an overall mark of at least 30% in the End-of-Unit Assessment.

Brief summary of the hurdle requirement	Rationale
To be eligible to obtain a pass in this unit, students must achieve an overall mark of at least 30% in the End-of-Unit Assessment.	In accordance with external accreditation requirements, the School of Engineering incorporates a hurdle requirement in order to achieve the Unit Learning Outcomes (ULOs) and Course Learning Outcomes (CLOs) of the course.

	A hurdle requirement is placed on the End-of-Unit Assessment component in order to meet the learning outcomes. Students are therefore required to achieve at least 30% in the End-of-Unit Assessment to pass this unit.

Summative assessment (tasks that will be graded or marked)
NOTE: It is *your responsibility* to keep a backup copy of every assignment and the materials used to develop/complete it where possible (e.g. written/digital reports, essays, videos, images). In the unusual event that one of your submissions becomes corrupted, is incorrectly submitted or otherwise lost, you may be asked to submit the backup copy. Any work you submit may be checked by electronic or other means for the purposes of detecting breaches of academic integrity such as collusion, plagiarism and contract cheating. You must understand your responsibility to act with honesty and integrity in your studies as Deakin takes all breaches very seriously. Make sure you read Your rights and responsibilities as a student in this unit to find out more about academic integrity.

Deakin has a universal assessment submission time of 8 pm AEDT/AEST. A late penalty will apply to assessments submitted after 11.59 pm AEDT/AEST.
Summative assessment task 1

Title	**Quiz 1**
Brief description of assessment task	Demonstrates an understanding in signalling and communication standards.
Detail of student output	Students are assessed through an online quiz with the technical knowledge acquired from the relevant topics (Unit Site quiz environment). Multiple choice, short answers matching type 14 questions in total with a time duration of one hour duration. This is an individual assessment task.
Grading and Weighting (% total mark for unit)	10% marked and graded
This task assesses your achievement of these Unit Learning Outcome(s)	ULO1, ULO5 Students will be required to answer questions regarding the necessary hardware and software environment and TCP/IP protocol architecture in this assessment.
This task assesses your achievement of these Graduate Learning Outcome(s)	GLO1, GLO5 The quiz will consist of questions relevant to the discipline knowledge at this level and questions stemming from real world examples underpinning the area of study.
How and when you will receive feedback on your work	The correct answers will be displayed after completing the Quiz online (via the unit site). Students may request individual feedback from the Unit Chair on their performance in the quiz within 10 University working days from receiving their mark.
When and how to submit your work	Quiz will be available **from 9am (AEDT) Monday 3 March 2025** (week 1) **until 5pm (AEDT) Monday 31 March 2025** (week 5). Students to complete and submit the Quiz via the unit site.

Summative assessment task 2

Title	**Quiz 2**
Brief description of assessment task	Demonstrates an understanding in networking protocols.
Detail of student output	Students are assessed through an online quiz with the technical knowledge acquired from the relevant topics (Unit Site quiz environment). Multiple choice, short answers matching type 12 questions in total with a time duration of one hour. This is an individual assessment task.
Grading and weighting (% total mark for unit)	10% marked and graded

This task assesses your achievement of these Unit Learning Outcome(s)	ULO2, ULO5 Students will be tested on their ability to answer questions regarding the need for protocols and networking principals and their relevance to TCP/IP protocol
This task assesses your achievement of these Graduate Learning Outcome(s)	GLO1, GLO5 The quiz will consist of questions relevant to the discipline knowledge at this level and questions stemming from real world examples underpinning the area of study.
How and when you will receive feedback on your work	The correct answers will be displayed after completing the Quiz online (via the Unit site). Students may request individual feedback from the Unit Chair on their performance in the quiz within 10 University working days from receiving their mark.
When and how to submit your work	Quiz will be available **from 9am (AEDT) Monday 3 March 2025** (week 1) **until 8pm (AEST) Monday 5 May 2025** (week 9). Students are required to complete and submit the quiz via the unit site.

Summative assessment task 3

Title	**Laboratory report**
Brief description of assessment task	Designing and implementing a data acquisition and communications.
Detail of student output	This is an individual assessment task. Students are required to apply the knowledge acquired from the relevant topics and solve appropriate problems and present the outcomes in a written form. The assignment will also contain some laboratory work designed to simulate routing algorithms, as well as exercise to identify frequency of actual acoustic signals and online training exercise concerning wireless communication where the results are expected to be presented as a part of the report for the individual assessment.
Grading and weighting (% total mark for unit)	30% marked and graded
This task assesses your achievement of these Unit Learning Outcome(s)	ULO3, ULO5 - Various type of networks and relation to TCP/IP architectures tested in a practical and applied sense. The application of the underlying principals is tested with computer simulations
This task assesses your achievement of these Graduate Learning Outcome(s)	GLO1, GLO2, GLO3, GLO4, GLO5, GLO6 - Technology use, conveying the technical and subject specific knowledge effectively is tested. The underlying questions are expected to be critically analysed, and solved using specific problem solving techniques in a timely fashion.
How and when you will receive feedback on your work	Students will be guided through the practical lectures and seminars during the relevant weeks/times.
When and how to submit your work	A report submitted through the unit site by **8pm (AEST), Friday 16 May 2025** (Week 10).

Summative assessment task End-of-unit assessment or Examination

Title	**End-of-Unit Assessment - written**
Brief description of assessment task	This is an individual assessment task. The task will assess and validate student knowledge of and ability to apply critical thinking techniques to identify and analyse problems from technical and non-technical perspectives. Students are expected to provide solutions to the questions with written answers uploaded within the specified time. Questions will be based on the topics covered in lecture during the entire trimester of study.

How you demonstrate your achievement	2-hour written assessment. A written assessment is a form of assessment that asks you to use text to communicate your knowledge, skill or understanding of a topic. It is completed online. At a set release time, you will be given the assessment instructions and will complete the assessment by a set deadline.
Grading and weighting (% total mark for unit)	50%
This task assesses your achievement of these Unit Learning Outcome(s)	ULO1, ULO2, ULO3, ULO4, ULO5 - The fundamental knowledge and in-depth understanding of the subjected matter is tested. The TCP/IP connection in a fundamental form is tested.
This task assesses your achievement of these Graduate Learning Outcome(s)	GLO1, GLO2, GLO4, GLO5, GLO6 - The understanding of fundamental principals, engage in critical thinking, use of problem solving philosophies are tested in a cohesive form across different topics of interest.
How and when you will receive feedback on your work	Information about the type of questions will be provided towards the end of the trimester. Students are encouraged to seek feedback during lecture as required. Deakin University will release the final assessment results at the stipulated timeframe. Students will receive a mark, which is an indicator of their overall performance in this unit of study.
When and how to submit your work	Students will be required to undertake a timed online assessment during the **End-of-Unit Assessment period**. You will have a start time (e.g., 9am) with a specified 2-hour time limit. Further details will be provided on the unit site. It is the responsibility of students to review their End-of-unit assessment timetable when it is released via StudentConnect. Further information including timetable release dates is available at end-of-unit assessments and examinations.

Your learning experiences in this unit

Educator-facilitated (scheduled) learning activities - on-campus unit enrolment

1 x 1 hour seminar per week, 3 x 1 hour practical experiences (laboratories) per trimester.
Educator-facilitated (scheduled) learning activities - online unit enrolment

Online independent and collaborative learning including a 1 x 1 hour seminar per week. Students are required to attend and participate in practical experiences (laboratories) at the Waurn Ponds (Geelong) Campus for the scheduled sessions during the trimester intensive activities as detailed in the unit site.
Typical study commitment

Students will on average spend 150 hours over the teaching period undertaking the teaching, learning and assessment activities for this unit.
This will include educator guided online learning activities within the unit site.
Teaching, learning, practical, and assessment activities for the unit for campus students includes seminar time, practicals, designated activities, assessment tasks, readings and study time. For online students the time should be divided between online learning activities, seminars, practicals at the Waurn Ponds campus during intensive week, discussion boards, designated activities, assessment tasks, readings and study time.

All the unit material/quizzes/assignment submission/messages will be via the unit site. The seminars will be used to interact with the students while communicating the subject matter.

Students may conduct literature searches and rely on external sources as an alternate learning experience. It is the

responsibility of the student to ensure the accuracy and authenticity of the material extracted from other sources – particularly certain Internet publications.

Note

At Deakin, courses are delivered within a learning environment that provides all students with equitable and consistent access to facilities, infrastructure, resources, and support to assist student progress and achievement of learning outcomes.

We have introduced new terms to reflect learning activities to enhance your learning experience, aligning with our innovative **DeakinDesign learning principles and practices**. The new terms better reflect how teaching teams will guide you through your learning journey and the types of learning experiences you will have.

'Lectures' are the activities where teaching staff engage you through presentations with student participation.

In **'seminars'**, an educator will guide you in a smaller group of students through highly interactive discussions and activities.

Your units may also include '**practical experiences**' such as '**laboratory**', '**workshops**', '**clinical skills**' and more. These hands-on activities typically take place in specialised facilities with industry tools, equipment or technology to allow you to apply your knowledge practically.

Some other terms

If you see a '**meeting**' in your timetable, this is an optional drop-in session.

'Assessments' or **'team-based learning**' indicate an in-class evaluation of your skills or knowledge. A **'pre-assessment practice'** could be scheduled to prepare you for these assessments.

Find out more

Take a look at the **Learning activities webpage** for a full list of the terminology changes and reasons they were changed.

Unit learning resources

Your unit learning resources can be accessed from your unit site.

Prescribed text(s): Stallings, 2013, Data and Computer Communications, 10th Ed, Pearson Education, Harlow, United Kingdom.

The texts and reading list for SEE312 can be found via the University Library.

Note: Select the relevant trimester reading list. Please note that a future teaching period's reading list may not be available until a month prior to the start of that teaching period so you may wish to use the relevant trimester's prior year reading list as a guide only.

Essential learning resources

Please note that we use the 8th, 9th or 10th edition of the prescribed text.

Other essential learning resources will be provided on the unit site.

Recommended learning resources

The Deakin Software Library provides students with access to software that you may need or find useful for your study at Deakin.

Where to access unit resources

Textbooks can be sourced from various outlets including direct from the publisher, online bookshops, or retailers. Limited copies of textbooks are also available on loan from the University Library.

Key dates for this study period

Trimester 1 teaching period begins	Monday 3 March 2025
Census date	Monday 31 March 2025
Easter/intra-trimester break	Friday 18 April - Sunday 27 April 2025
Trimester 1 teaching period ends	Friday 23 May 2025
Study period (end-of-unit assessment /examination preparation period)	Monday 26 May - Friday 30 May 2025
End-of-unit assessment and examination period	Monday 2 June - Friday 13 June 2025
Inter-trimester break (the period between trimesters)	Monday 16 June - Friday 4 July 2025
Unit results released	Thursday 3 July 2025 (10.30 am)

Unit weekly activities

Week	Commencing	Topic	Assessment due date
1	3 March 2025	Introduction to Data Communications (CH 1 & 2)	
2#	10 March 2025	Data Transmission (CH 3 & 4)	
3	17 March 2025	Data Transmission (CH 3 & 4)	
4	24 March 2025	Signal Encoding (CH 5)	
5	31 March 2025	Signal Encoding and Data Transmission Techniques (CH 5 & CH 6)	**Assignment 1: Quiz** closes 31 March
6	7 April 2025	Data Transmission Techniques (CH 6)	
7*~	14 April 2025	Introduction to Multiplexing (CH 8)	
8	28 April 2025	WAN Technology and Protocols (CH 9)	
9	5 May 2025	Cellular Wireless Networks (CH 10)	**Assignment 2: Quiz** closes 5 May
10	12 May 2025	Routing in Switched networks (CH19)	**Assessment 3: Laboratory Report** due 16 May
11	19 May 2025	Local Area Networks (LANS) (CH11), Revision	
12	26 May 2025		Study period
13-14^	2 June 2025		End-of-Unit Assessment and exam period

Labour Day public holiday: **Monday 10 March 2025** - University closed
* Easter/Intra-trimester break: **Friday 18 April - Sunday 27 April 2025** (between weeks 7 and 8)
~ ANZAC Day public holiday: **Friday 25 April 2025** - University closed
^ King's Birthday public holiday: **Monday 9 June 2025** - University closed